

```
ubuntu@devops:~$ ssh -i ~/.ssh/id_rsa ubuntu@slave1
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1065-gcp x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

```
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.
```

```
https://ubuntu.com/engage/secure-kubernetes-at-the-edge
```

This system has been minimized by removing packages and content that are not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

2 additional security updates can be applied with ESM Apps.

Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

The programs included with the Ubuntu system are free software; the exact distribution terms for each program are described in the individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.

```
ubuntu@slave1:~$ cat .ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQC5gqexgl8ylMNOASKjT589h60zXHoY3um3LyBEqD04xi1vWAv2f+vFW2QWQMSkLm19GHMffavigY4TUTrXF7Y4I+ra9WFXjScWZul
EY6HyUAH3yvA5TSfKQHZkCfgr61N0IlQ0yxKpi6+eGdGjorDNjFITAuAvw3tMILjkcX2qfx6MqWF4Ouil7CNTn8KN1PZpt/tLaz7neCHXBO4NXGYmFlqci3ArNsbD4qAO3CISMfii
ub8gkwlhIKNj9hbYxIAiq4GApe/2sMPmK4uRjHfLmfnVy8Tc+Ene3SrGRcmB8X4Av9DWyBFKqQkQ75rNLQrs2b6BLv3H7Wk86ChXtTq/S71YfukTYDxTdlHzJGEBglxXQlVBCQWUo1
D0A/tD/YqBZ8zYEBEBqssSRytIGBU3oJa4klNx4jVIFc8BsLzbiQDUAEvJbAiGrJY3xuiFqQJVAiW6uSFhBu6yM76Cm26f8d7Fi6n/fX8XUPpAUKegwqE5R5ZvD2sueT46Q98= ubu
ntu@devops
ubuntu@slave1:~$ ls -ld ~/.ssh
drwx----- 2 ubuntu ubuntu 4096 Aug 11 15:54 /home/ubuntu/.ssh
ubuntu@slave1:~$ ls -l ~/.ssh/authorized_keys
-rw-rw-r-- 1 ubuntu ubuntu 567 Aug 11 15:46 /home/ubuntu/.ssh/authorized_keys
ubuntu@slave1:~$ stat -c '%U %G %a %n' ~/.ssh ~/.ssh/authorized_keys
ubuntu ubuntu 700 /home/ubuntu/.ssh
ubuntu ubuntu 664 /home/ubuntu/.ssh/authorized_keys
ubuntu@slave1:~$ chmod 600 ~/.ssh/authorized_keys
ubuntu@slave1:~$ chown -R ubuntu:ubuntu ~/.ssh
ubuntu@slave1:~$
```

Activate Windows
Go to Settings to activate Windows.

```
ubuntu@devops:~$ ansible -m ping devops
[WARNING]: Platform linux on host devops is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of
another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
devops | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.10"
  },
  "changed": false,
  "ping": "pong"
}
ubuntu@devops:~$ ansible -m ping slave1
[WARNING]: Platform linux on host slave1 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of
another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
slave1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.10"
  },
  "changed": false,
  "ping": "pong"
}
ubuntu@devops:~$
```

Activate Windows
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```
ubuntu@devops:~$ ansible -m ping webserver
[WARNING]: Platform linux on host devops is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
devops | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.10"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Platform linux on host slave1 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
slave1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.10"
  },
  "changed": false,
  "ping": "pong"
}
ubuntu@devops:~$
```

Activate Windows
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```
ubuntu@devops:~$ ansible -m ping slave1
[WARNING]: Platform linux on host slave1 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of
another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
slave1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.10"
  },
  "changed": false,
  "ping": "pong"
}
ubuntu@devops:~$ ansible -m setup slave1
[WARNING]: Platform linux on host slave1 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of
another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
slave1 | SUCCESS => {
  "ansible_facts": {
    "ansible_all_ipv4_addresses": [
      "10.128.0.5",
      "172.17.0.1"
    ],
    "ansible_all_ipv6_addresses": [
      "fe80::4001:aff:fe80:5",
      "fe80::5cc3:78ff:fe54:fdd5"
    ],
    "ansible_apparmor": {
      "status": "enabled"
    },
    "ansible_architecture": "x86_64",
    "ansible_bios_date": "07/24/2026",
    "ansible_bios_vendor": "Google",
    "ansible_bios_version": "Google",
    "ansible_board_asset_tag": "5A8E444C-1758-D499-A252-64359C149C27",
    "ansible_board_name": "Google Compute Engine",
```



```
    ],
    "ansible_system_capabilities_enforced": "True",
    "ansible_system_vendor": "Google",
    "ansible_uptime_seconds": 271033,
    "ansible_user_dir": "/home/ubuntu",
    "ansible_user_gecos": "Ubuntu",
    "ansible_user_gid": 1000,
    "ansible_user_id": "ubuntu",
    "ansible_user_shell": "/bin/bash",
    "ansible_user_uid": 1000,
    "ansible_userspace_architecture": "x86_64",
    "ansible_userspace_bits": "64",
    "ansible_virtualization_role": "guest",
    "ansible_virtualization_tech_guest": [
        "kvm"
    ],
    "ansible_virtualization_tech_host": [],
    "ansible_virtualization_type": "kvm",
    "discovered_interpreter_python": "/usr/bin/python3.10",
    "gather_subset": [
        "all"
    ],
    "module_setup": true
  },
  "changed": false
}
```

```
ubuntu@devops:~$ ansible -m command -a "hostname" slave1
```

```
[WARNING]: Platform linux on host slave1 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
```

```
slave1 | CHANGED | rc=0 >>
```

```
slave1
```

```
ubuntu@devops:~$
```

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core/2.17/reference_appendices/interpreter_discovery.html for more information.

slave1 | CHANGED | rc=0 >>

slave1

ubuntu@devops:~\$ vim first_playbook.yaml

ubuntu@devops:~\$ cat first_playbook.yaml

- hosts: slave1

become: yes

name: Play 1

tasks:

- name: Execute command 'Date'

command: date

- name: Install nginx

apt: name=nginx state=latest

ubuntu@devops:~\$ ansible-playbook first_playbook.yaml

PLAY [Play 1] *****

TASK [Gathering Facts] *****

[WARNING]: Platform linux on host slave1 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.

ok: [slave1]

TASK [Execute command 'Date'] *****

changed: [slave1]

TASK [Install nginx] *****

changed: [slave1]

PLAY RECAP *****

slave1 : ok=3 changed=2 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

ubuntu@devops:~\$

Activate Windows
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TASK [Install nginx] *****
changed: [slave1]

PLAY RECAP *****
slave1 : ok=3 changed=2 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

```
ubuntu@devops:~$ ansible-galaxy init first_role
- Role first_role was created successfully
ubuntu@devops:~$ sudo apt install tree
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  tree
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 47.9 kB of archives.
After this operation, 116 kB of additional disk space will be used.
Get:1 http://us-central1.gce.archive.ubuntu.com/ubuntu jammy/universe amd64 tree amd64 2.0.2-1 [47.9 kB]
Fetched 47.9 kB in 0s (1587 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package tree.
(Reading database ... 133287 files and directories currently installed.)
Preparing to unpack .../tree_2.0.2-1_amd64.deb ...
Unpacking tree (2.0.2-1) ...
Setting up tree (2.0.2-1) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dia
log.pm line 78.)
debconf: falling back to frontend: Readline
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.
```

Activate Windows
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SSH-in-browser

```
debconf: falling back to frontend: Readline
Scanning processes...
Scanning linux images...
```

```
Running kernel seems to be up-to-date.
```

```
No services need to be restarted.
```

```
No containers need to be restarted.
```

```
No user sessions are running outdated binaries.
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

```
ubuntu@devops:~$ tree first_role
```

```
first_role
```

```
├── README.md
├── defaults
│   └── main.yml
├── files
├── handlers
│   └── main.yml
├── meta
│   └── main.yml
├── tasks
│   └── main.yml
├── templates
├── tests
│   ├── inventory
│   └── test.yml
└── vars
    └── main.yml
```

```
8 directories, 8 files
```

```
ubuntu@devops:~$
```

8 directories, 8 files

```
ubuntu@devops:~$ vim first_role/tasks/main.yml
```

```
ubuntu@devops:~$ cat first_role/tasks/main.yml
```

```
---
```

```
# tasks file for first_role
```

```
# ---
```

```
# tasks file for first_role
```

```
- include: install.yml
```

```
ubuntu@devops:~$ vim first_role/tasks/install.yml
```

```
ubuntu@devops:~$ cat first_role/tasks/install.yml
```

```
---
```

```
- name: Execute command 'Date'
```

```
  command: date
```

```
  register: date_output
```

```
- name: Print date
```

```
  debug:
```

```
    msg: "{{date_output}}"
```

```
- name: Install nginx
```

```
  apt: name=nginx state=latest
```

```
ubuntu@devops:~$
```



```
TASK [Gathering Facts] *****
[WARNING]: Platform linux on host slave1 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of
another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
```

```
ok: [slave1]
```

```
TASK [first_role : include_tasks] *****
included: /home/ubuntu/first_role/tasks/install.yml for slave1
```

```
TASK [first_role : Execute command 'Date'] *****
changed: [slave1]
```

```
TASK [first_role : Print date] *****
```

```
ok: [slave1] => {
```

```
  "msg": {
    "changed": true,
    "cmd": [
      "date"
    ],
    "delta": "0:00:00.005529",
    "end": "2026-08-11 18:12:02.255824",
    "failed": false,
    "msg": "",
    "rc": 0,
    "start": "2026-08-11 18:12:02.250295",
    "stderr": "",
    "stderr_lines": [],
    "stdout": "Tue Aug 11 18:12:02 UTC 2026",
    "stdout_lines": [
      "Tue Aug 11 18:12:02 UTC 2026"
    ]
  }
}
```

```
TASK [first_role : Install nginx] *****
ok: [slave1]
```

```
PLAY RECAP *****
slave1                : ok=5    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
ubuntu@devops:~$
```